

Simulation Plan

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August 17, 2026

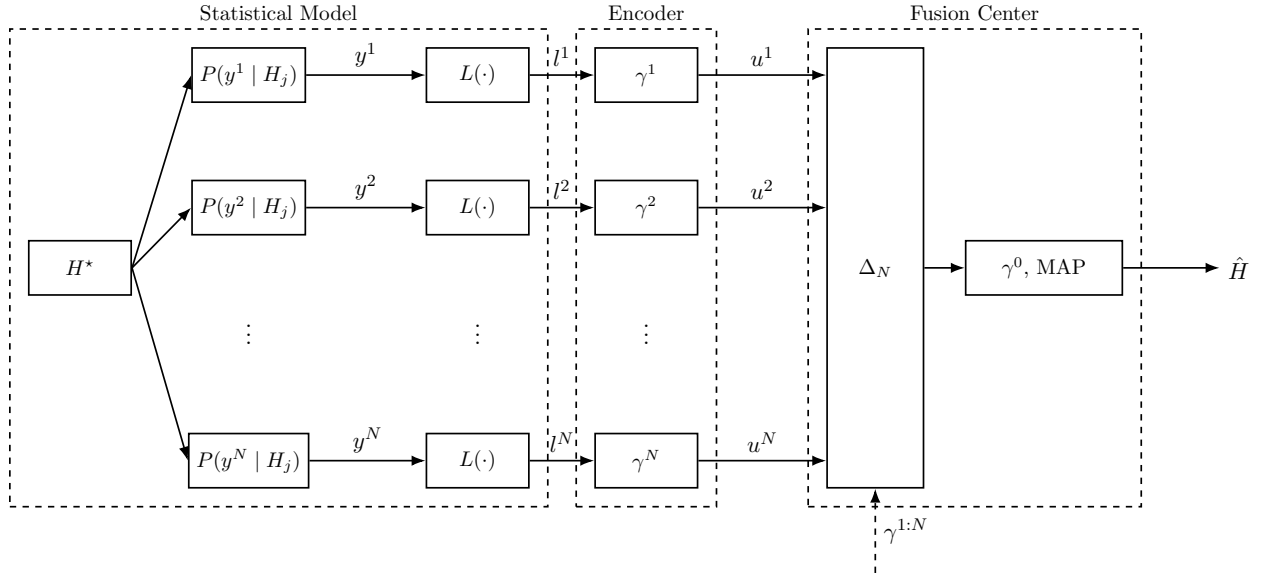


Figure 1: Block diagram of the decentralized detection system. The source H^* generates the hypothesis observed by N sensors as $y^i | H_j$. Each sensor reduces its observation to the likelihood ratio $l^i = L(y^i)$, which is a sufficient statistic for H , and quantizes it via the threshold policy γ^i into a message u^i . The fusion center forms the statistic Δ_N from all messages together with the realized encoding policies $\gamma^{1:N}$ (dashed), and applies the MAP rule γ^0 to produce the decision $\hat{H}$.

1 Simulation 1 - One bit per sensor

In this simulation, each sensor can transmit up to 1 bit, encoding its local decision as u^1 or u^2 . We evaluate the Normalized Error Exponent $J_{EE}^N = -\frac{1}{N} \log(J^N)$ as a function of N , the number of sensors observing the environment. The statistical model is Gaussian, with the mean varying across hypotheses, and all sensors are assumed to share the same policy.

Setup A - Always Transmit Policy (Vanilla)

Each sensor observes $y \sim \mathcal{N}(0, \sigma^2)$ under H_1 or $y \sim \mathcal{N}(\mu, \sigma^2)$ under H_2 , and always transmits one bit indicating which side of a fixed threshold its observation falls on. The threshold is set at $\tau = \frac{\mu}{2}$, the midpoint between the two hypothesis means.

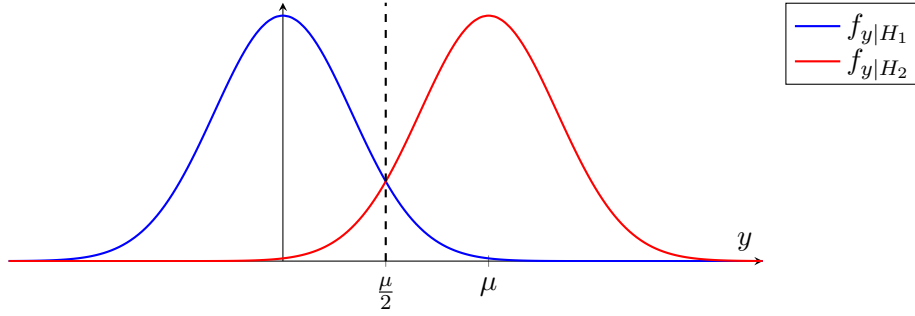


Figure 2: Gaussian model for Setup A. A sensor decides u^1 if $y < \mu/2$ and u^2 if $y \geq \mu/2$, and always transmits its 1-bit decision.

Setup B - Silent Policy

Each sensor observes the same Gaussian model as in Setup A, but now transmits its decision only when confident about it: it sends u^1 if $y < \lambda_1$, u^2 if $y \geq \lambda_2$, and stays silent otherwise. The two thresholds $\lambda_1 < \lambda_2$ are placed symmetrically around the midpoint $\mu/2$, so a sensor only transmits when its observation falls far enough from the boundary between the hypotheses.

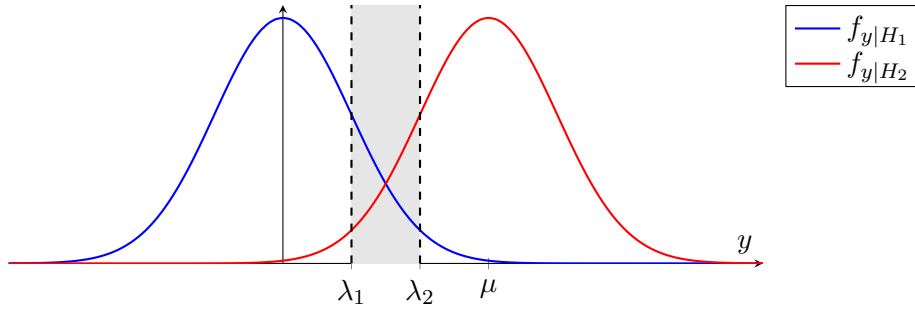


Figure 3: Gaussian model for Setup B. A sensor sends u^1 if $y < \lambda_1$, u^2 if $y \geq \lambda_2$, and stays silent (shaded region) otherwise.

2 Simulation 2 - R bits per sensor

In this simulation, each sensor can transmit up to $R = \log_2(L)$ bits, encoding its local decision into $\mathcal{U} = \{u^1, u^2, \dots, u^L\}$. We will fix the number of sensors $N =$ and evaluate the Normalized Error Exponent $J_{EE}^N = -\frac{1}{N} \log(J^N)$ as a function of R , the number of bits allocated per sensor. We will note how many sensors were transmitting at each simulation. The statistical model is Gaussian, with the mean varying across hypotheses, and all sensors are assumed to share the same symmetric and evenly distributed policy.

Setup A - Always Transmit Policy (Vanilla)

Each sensor observes $y \sim \mathcal{N}(0, \sigma^2)$ under H_1 or $y \sim \mathcal{N}(\mu, \sigma^2)$ under H_2 , and always transmits R bits indicating which hypothesis it observed and how strong was the observation.

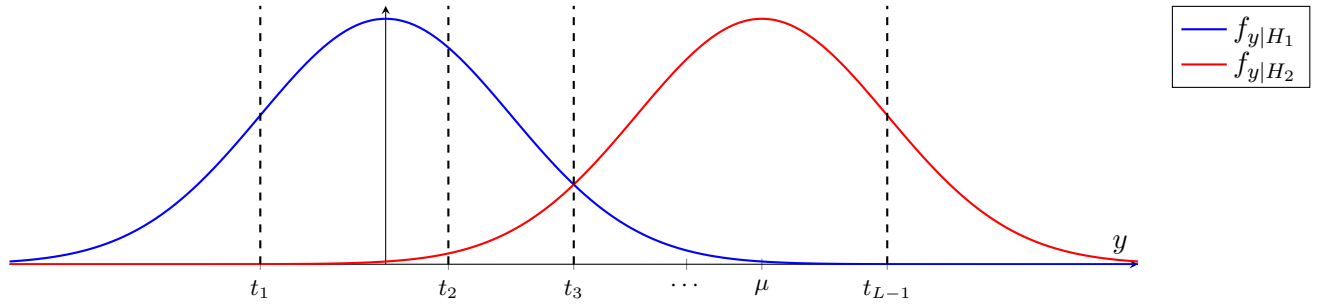


Figure 4: Gaussian model for Setup A under Simulation 2. The observation space is partitioned by $L - 1$ thresholds $t_1 < t_2 < \dots < t_{L-1}$ into L regions, each mapped to a distinct symbol $u^i \in \mathcal{U}$; the thresholds extend beyond both means, and the sensor always transmits the $R = \log_2(L)$ bits identifying its region.

Setup B - Silent policy

Each sensor observes the same Gaussian model as in Setup A, but now stays silent when its observation is inconclusive. The observation space is split using L thresholds (one more than in Setup A) into $L + 1$ regions: $L/2$ bins on each side of a central *silence bin*. The silence bin is symmetric about the midpoint $\mu/2$ with half-width δ , bounded by $\mu/2 - \delta$ and $\mu/2 + \delta$. A sensor stays silent if its observation falls inside the silence bin, and otherwise always transmits the $R = \log_2(L)$ bits identifying which of the remaining L regions its observation fell into.

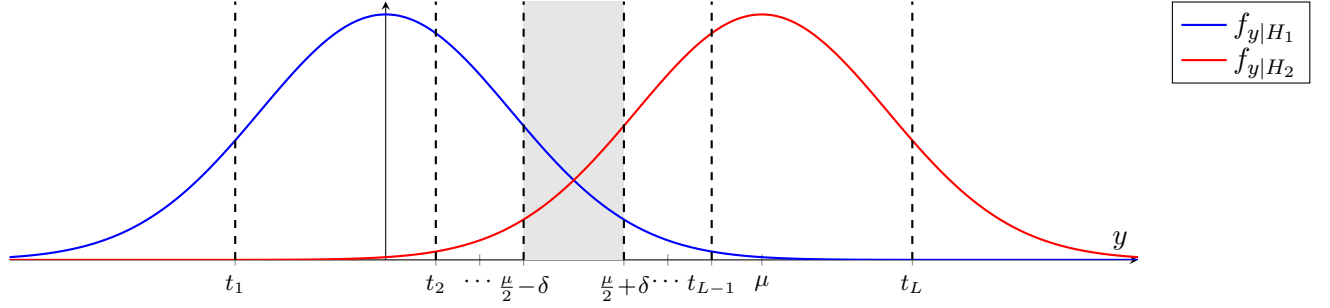


Figure 5: Gaussian model for Setup B under Simulation 2. The observation space is partitioned by L thresholds into $L/2$ bins on each side of the shaded silence bin, which is centered on $\mu/2$ with half-width δ ; a sensor stays silent if its observation falls in the shaded region, and otherwise transmits the $R = \log_2(L)$ bits identifying its region.